Technology: implementing observability

Categorizing observations

Three pillars of observability are logs, traces, and metrics.

Reporting observations:

- visually/graphically
- textually
- verbally
- quantitative and qualitative.

Assign observations to the RACI individuals:

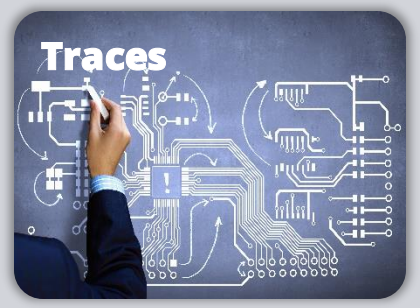
- Who decides on action?
- If no action, downgrade the metric.

Scale metrics:

- Use 1-N on importance.
- Know when and why to respond.



Time stamps support investigation of events from one point in time to another.



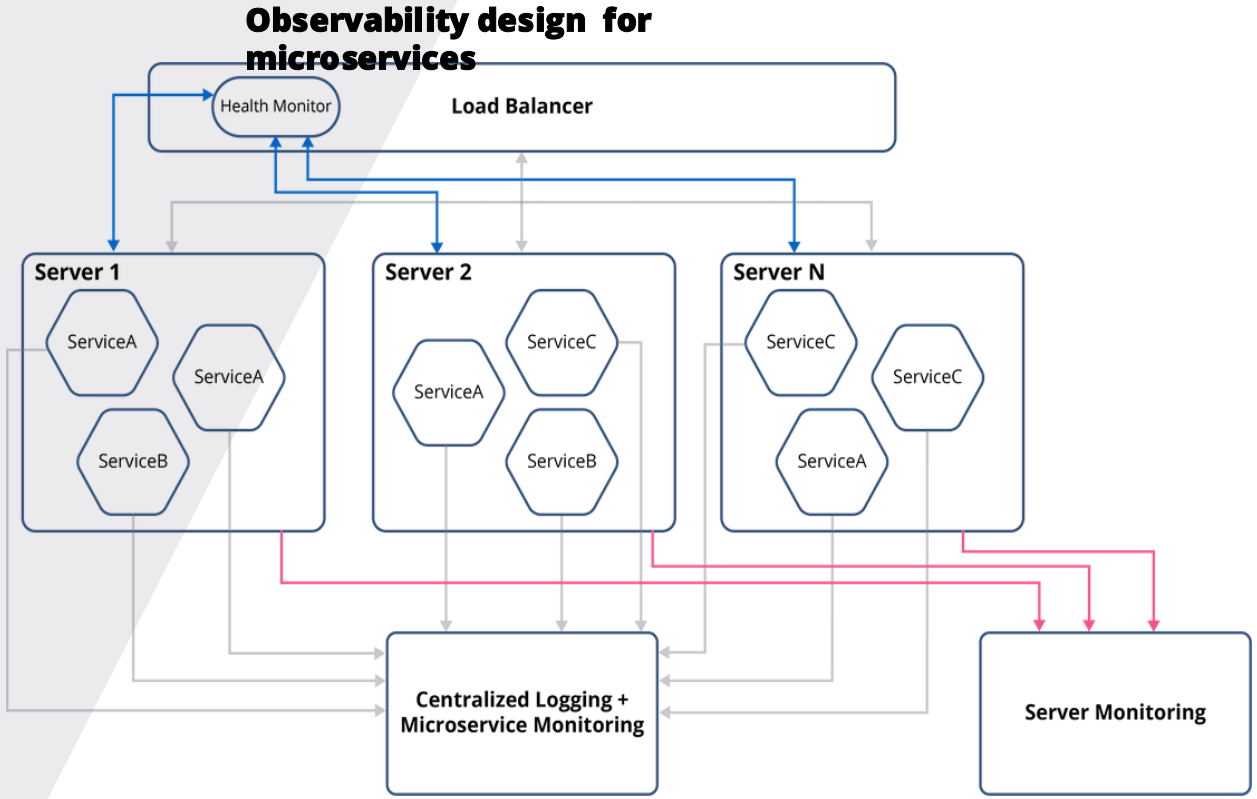
Events are traceable across a distributed system.



Aggregate and organize data according to algorithms.

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Health checks and heartbeats



Source: https://www.amazon.com/Designing-Microservices-Platforms-NATS-microservices/dp/1801072213

All observability should require authentication:

- can use correlation ID to verify repeated asks
- can drive logs to central locations.

Health check; reach out and make sure a tool is running.

Heartbeat; send a regular signal from the running application.

Microservice observability:

- server monitoring; CPU below 70%
- application monitoring
- log monitoring
- health monitoring.

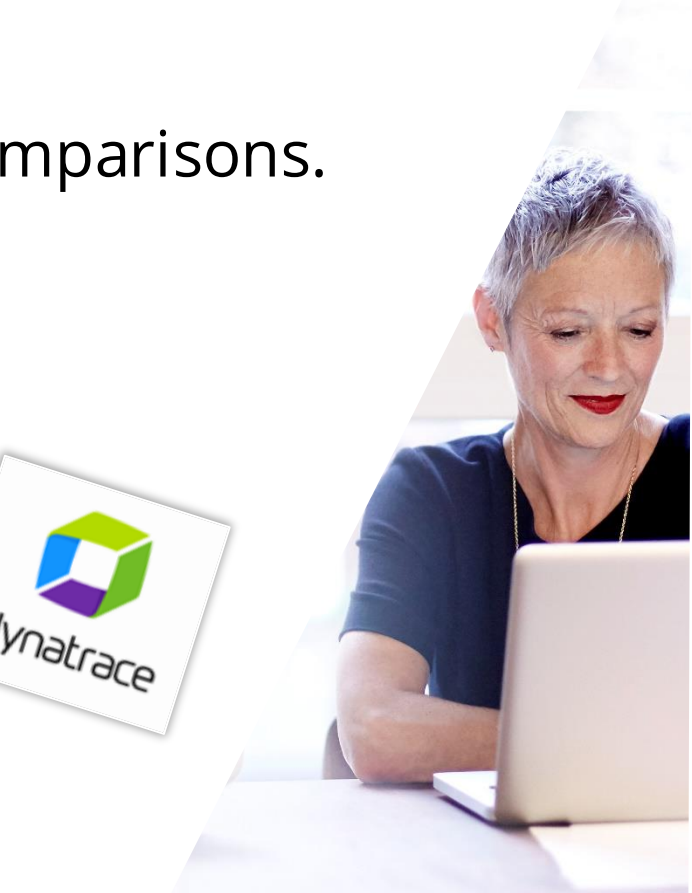
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Working with observability tool vendors

Forrester and Gartner reports are a good source of tool comparisons.

- Tools are based on requirements.
- Need to investigate multiple vendors.
- Most vendors offer evaluation calls:
  - Initial call: develop requirements.
  - Second call: build a model and generic demo.
  - Third call: tailored demo.
- Some tools offer free trial periods.





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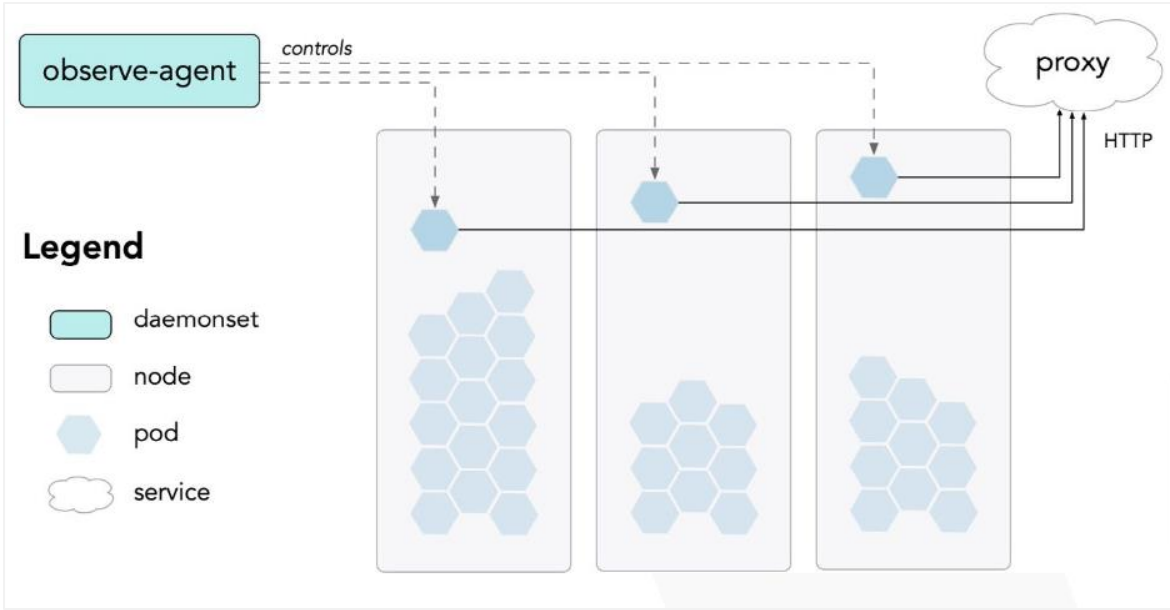
Observing Kubernetes

- Kubernetes contains multiple distinct services for infrastructure and containers.
- No single set of logs, metrics or traces exists.
- Each cluster can be interpreted separately:
  - no centralized application data
  - rapidly changing clusters
  - must export data to stable environment to observe.
- Must have context, history, and overall environment state.

Keys to K8s

- Do not just aggregate logs.
- Do not focus on metrics alone.
- Do not focus just on applications.
- Do not rely on a managed Kubernetes service.





Source: https://www.observeinc.com/resources/observability-in-kubernetes/

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